

ELITE

for the TRS-80 Color Computer (CoCo 1 / 2)

User's Manual



A from-scratch 6809 rewrite of the BBC Micro classic by Ian Bell and David Braben.
Same galaxy, same rules, same ships, drawn in wireframe on the CoCo's 256 x 192 screen, with the original's dashboard and its red scanner.

1. Introduction

You are Commander Jameson, a beginner in a Cobra Mk III trading ship, docked at the planet Lave with 100.0 credits, a pulse laser and a little fuel. Eight galaxies of 256 star systems wait for you. How you earn your living is your own business: trade goods between worlds, hunt pirates for bounty, mine asteroids, smuggle, or take on the special missions. Your *rating* climbs from Harmless through Mostly Harmless, Poor, Average, Above Average, Competent, Dangerous and Deadly to ELITE with every kill you make.

This version follows the original's data and rules: galaxy generation, market prices, ship designs, enemy behaviour, missions and descriptions all come from the 6502 original. The game runs at about four frames a second. That is the pace of the CoCo, not a fault of your machine.

2. What you need

- A CoCo 1 or 2 with 64K RAM, or an emulator (the game was developed with XRoar 1.6).
- Either the disk version (ELITE.DAT + ELITE.BIN on a Disk BASIC 1.0/1.1 disk, with room for the save file ELITE.SAV), or the cartridge version (a banked ROM of the Super Program Pak kind). The cartridge has no room to save commanders.
- The keyboard. There is no joystick support; the arrow keys fly the ship.

3. Starting the game

Disk: insert the disk and type

CLEAR 200,&H2800:LOADM"ELITE":EXEC

The program loads its overlays (ships, docked screens) from the disk as it needs them, so each change between flying and the station takes some seconds. **Cartridge:** switch on, the game starts by itself.

The title screen

The title shows the ships of the game one after another, flying towards you.

Key	What it does
LEFT / RIGHT	Show the next / previous ship
T	Show or hide the ship's name
SPACE	Pause
ENTER or N	Start a new game with the default commander (Jameson, docked at Lave)
Y	Load a saved commander from the disk first

4. The keyboard at a glance

Key	In flight	Docked
Arrow keys	Roll and pitch	Move the crosshairs on a chart (SHIFT: four times as far)
SPACE	Speed up	
/	Slow down	
0	Front view	Launch
1, 2, 3	Rear, left, right view	1: buy goods, 2: sell goods, 3: equip ship
4	Long-range chart	Long-range chart
5	Short-range chart	Short-range chart
6	Data on the selected system	Data on the selected system

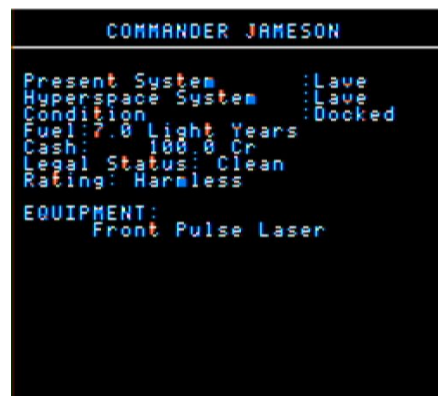
7	Market prices	Market prices
8	Status	Status
9	Inventory	Inventory
A	Fire laser	
T / U / M	Target missile / unarm / fire	
E	E.C.M.	
B	Energy bomb	
BREAK	Escape pod	
C / P	Docking computer on / off	
H	Hyperspace jump	
G	Galactic hyperdrive	
Q	Sound off / on	Sound off / on
O, D, F		On a chart: crosshairs home; name the nearest system; find a system by name
@		Disk menu (disk version only)

In flight, the screens 4 to 9 pause the game while you look at them; press 0 to 3 to return to a space view.

5. In the station

You start docked. All business is done from the station's screens. Press a number key to change screen.

Status (8)



Your name, present and hyperspace systems, condition (Docked, Green, Yellow, Red), fuel, cash, legal status (Clean, Offender, Fugitive), rating and equipment.

Market prices (7), buying (1) and selling (2)

LAVE MARKET PRICES			
PRODUCT	UNIT	UNIT PRICE	QUANTITY FOR SALE
Food	t	4.0	17t
Textiles	t	20.0	16t
Radioactives	t	20.0	18t
Slaves	t	1.0	-
Liquor/Wine	t	1.0	21t
Luxuries	t	1.0	15t
Narcotics	t	1.0	-
Computers	t	1.0	11t
Machinery	t	1.0	13t
Alloys	t	1.0	-
Firearms	t	1.0	42t
Furs	t	1.0	53t
Minerals	t	1.0	-
Gold	kg	1.0	8kg
Platinum	kg	1.0	2kg
Gem-Stones	kg	1.0	-
Alien Items	t	51.0	54t

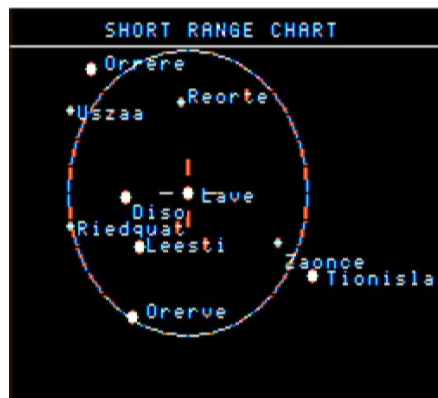
Every world has its own prices and stock. Buy where a good is cheap and sell where it is dear: that is the whole art of trading. Food, textiles and radioactives are cheap on agricultural worlds; computers, machinery and alloys on

industrial ones. Narcotics, firearms and slaves are illegal in some places and will make you an Offender when you carry them through a hostile system. The buy screen asks, item by item, how many units you want: type the number and press ENTER; ENTER alone skips the item, Y as the first key takes all you can afford and carry, N none. The sell screen does the same for the goods in your hold. Your hold is limited (20 t to begin with, 35 t with the large cargo bay), and gold, platinum and gem-stones count in kg and g rather than tonnes.

Equip ship (3)

Fuel, missiles, large cargo bay, E.C.M. system, extra pulse lasers, beam lasers, fuel scoops, escape pod, energy bomb, extra energy unit, docking computer, galactic hyperdrive and military or mining lasers, as the tech level of the system allows. Type the number of the item and press ENTER; ENTER alone leaves the screen. A laser goes on the front, rear, left or right: you choose the mount. A new laser replaces the old one on that mount and you get its value back.

Charts (4 and 5) and Data on system (6)



The **long-range chart** shows the whole galaxy; the **short-range chart** shows the systems within reach of your fuel, with a circle that marks the range of a jump. Move the crosshairs with the arrow keys (hold SHIFT for bigger steps). D names the system nearest to the crosshairs and its distance; F asks for a name and puts the crosshairs on that system; O brings them back to where you are. The system under the crosshairs is the *selected* system: screen 6 then tells you about it.



Data on system gives the economy, government, tech level, population (and the species), gross productivity and average radius, followed by a description. Dictatorships and anarchies tend to be dangerous; rich industrial worlds with high tech levels sell the best equipment.

Launching (0)

Press 0 to leave the station: you fly out through the slot, and you are in space.

6. Flying



The dashboard

Instrument	Meaning
FS, AS	Forward and aft shields
FU	Fuel (each jump uses fuel)
CT	Cabin temperature: rises near a sun; at the top the ship burns up. A fuel scoop lets you refuel from a sun, carefully
LT	Laser temperature: do not hold the trigger down for ever
AL	Altitude above the planet's surface
SP	Speed
RL, DC	Roll and pitch (dive and climb) indicators
1 to 4	The four energy banks. Energy runs your shields; when they are gone, you lose the ship
Compass	Points to the planet (or the station once you are near)
Scanner	The elliptical 3-D radar. Ships show as bars: above the plane or below; colours separate friends from foes. The missile indicators show your missiles (ready, armed, locked)
S, E (letters on the right)	Space station nearby, and E.C.M. status

Flying the ship

The arrow keys roll and pitch. SPACE accelerates and / slows down. Switch between the front, rear, left and right views with 0 to 3; your lasers fire in the direction of the view if you have one mounted there.

Combat

- **Laser (A):** fires the laser of the present view. Keep the target in the crosshairs.
- **Missiles:** T arms a missile and locks on a ship near the crosshairs; M fires it; U unarms it.
- **E.C.M. (E):** destroys missiles in flight, at a cost of energy. Needs the E.C.M. system.
- **Energy bomb (B):** destroys every ship near you, except the station, in one flash. It is used up.
- **Escape pod (BREAK):** launches you away from your ship and lands you back at the station of the system you are in. You lose the cargo; your legal status is cleared and the tank is full, but you keep your rating and equipment. Needs the escape pod, which is used up.

Killing pirates, thargoids and other hostile ships raises your rating; shooting innocents (traders, the police, the station) makes you an Offender or a Fugitive, and the police will turn on you.

Scooping and mining

With the fuel scoop fitted, cargo canisters, escape pods and the like that float *below* you are scooped up when you fly into them, if there is room in the hold; without room, or without a scoop, you crash into them. Near a sun

the scoop also fills the tank (watch the cabin temperature). Asteroids can be broken up with a mining laser. Slaves and narcotics you pick up may be illegal.

Warnings

The ship's computer prints messages at the bottom of the screen: INCOMING MISSILE, ENERGY LOW, DOCKING COMPUTERS ON, FUEL SCOOPS ON and so on. When your shields are down, a hit or a crash can destroy cargo or equipment ("... DESTROYED").

7. Travel

Hyperspace (H)

Open a chart (4 or 5), move the crosshairs to the system you want, then press H. If it is within range of your fuel, the countdown begins (15, 14, ... 1) and you jump. The fuel used equals the distance in light years (a full tank is 7.0). Beware of mis-jumps: a jump can occasionally go wrong and leave you in witchspace, where the only thing around is the thargoids' ships.

Galactic hyperdrive (G)

The galactic hyperdrive (an equipment item) takes you to the same position in the next galaxy (galaxy 1 to 8, then back to 1). The drive is used up by the jump.

Docking

Every system has a station orbiting its planet. Fly towards it; when you are near, the compass points at the station instead of the planet. To dock, point at the station, line up with the slot of the rotating station, match its rotation by rolling so that the slot is horizontal, and fly through the slot slowly. Hitting the station or approaching the wrong way is a collision: below speed 5 it only costs energy, at speed 5 or more it destroys your ship. If you have the **docking computer**, switch it on with C: it will fly the ship in for you (it takes about two minutes on this version), and you can turn it off at any moment with P. When you are docked, you pass through the station tunnel into the hangar.

8. Missions

Two missions of the original are in the game. Mission 1 is the hunt for the **Constrictor**, a deadly military prototype stolen from the Navy; you are briefed when you dock in one of the first two galaxies with a rating of Competent or better. Destroying the Constrictor completes the mission; docking again earns the reward (5000 credits). Mission 2 follows in the third galaxy once your rating is much higher: you are sent to a distant planet, receive the Thargoid plans there and must deliver them to another; the Thargoids will try to stop you. The briefings are shown on the station screens, the first with the rotating Constrictor.

9. Saving and loading (disk version)

At the title press Y to load a commander. In the station press @ for the disk menu: load a commander, save the present one, show the catalogue or delete a commander. Commanders are kept in the file ELITE.SAV on the disk (18 sectors). The name is typed in capitals; the left arrow deletes a letter and ENTER finishes. The cartridge version cannot save.

10. Sound

The CoCo's 6-bit DAC gives the laser, hits, explosions and alarms. Q switches sound off and on.

11. Hints for new commanders

- Start by trading cheap goods between the nearest systems. Look at the market (7) before every jump.
- Always keep enough fuel to come back; refuel at the station (screen 3).
- Buy a few missiles and an E.C.M. early; they are cheap compared with a new ship.
- Pirates in safer systems are weak: that is where to practise combat. Stay away from anarchies when carrying valuables.
- Do not fire at the station or at the police. If you become a Fugitive, you must stay out of civilised systems for a while.
- Docking computers are expensive but save your nerves; learn to dock by hand anyway.

12. Troubleshooting

Problem	Remedy
Game is slow	That is normal: about four frames a second. Do not run an emulator in unthrottled mode: the game's timing comes from the 60 Hz vertical sync.
A key press is lost	The keyboard is scanned 30 times a second, so a very short tap is still caught; hold the key a little longer on the title screen.
Disk errors on load	The program expects Disk BASIC 1.0 or 1.1 and reads its own sectors; check that the disk is not write-protected for saving.
Cannot save	The cartridge version has no room for the save file.